Client Change Request (CCR)

CCR Number: CCR-008

Client: Sunshine Elementary School

Submitted By: Lisa Hernandez, Fourth Grade Teacher

Priority: Medium

Requested Delivery: End of Today's Sprint

Assigned Developer: _____

Date Assigned: _____

Business Background

Sunshine Elementary School allows teachers to award extra credit to students who complete optional classroom activities. Currently, teachers record a student's original grade and then manually calculate the new grade after adding any extra credit points.

The current process requires teachers to repeatedly rewrite grades on paper, increasing the likelihood of transcription errors. The school would like a simple computer program that updates the student's original grade directly rather than creating multiple copies of the same information.

At this time, the school only needs to process one student's grade each time the program is executed.

Customer Requirements

The teacher should be able to enter:

- A student's original grade.
- The number of extra credit points to be awarded.

The program should:

- Update the student's original grade by adding the extra credit points.
- Display the student's original grade.
- Display the corrected grade after the extra credit has been applied.

The report should be clearly formatted so that the teacher can easily verify the results.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

☐ Teacher can enter a student's original grade.

☐ Teacher can enter the extra credit points.

☐ Program correctly updates the student's original grade.

☐ Program displays the original grade.

☐ Program displays the corrected grade.

☐ Program produces a neatly formatted report.

☐ Program completes without errors.

Out of Scope

The customer is **not** requesting:

- Processing multiple students
- Saving grades to a file
- Calculating class averages
- Letter grades (A, B, C, etc.)
- Grade reports
- Student names
- Multiple assignments
- Databases
- Printing report cards
- Graphical user interfaces (GUI)

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Teacher Input

Original Grade: 84

Extra Credit: 5

Expected Output

Sunshine Elementary School

Grade Adjustment Report

Original Grade : 84

Extra Credit : 5

Corrected Grade: 89

Example Scenario 2

Teacher Input

Original Grade: 92

Extra Credit: 3

Expected Output

Sunshine Elementary School

Grade Adjustment Report

Original Grade : 92

Extra Credit : 3

Corrected Grade: 95

Example Scenario 3

Teacher Input

Original Grade: 100

Extra Credit: 0

Expected Output

Sunshine Elementary School

Grade Adjustment Report

Original Grade : 100

Extra Credit : 0

Corrected Grade: 100

Notes from the Customer

"I don't want teachers rewriting grades throughout the program. There should be one official grade that gets updated whenever extra credit is awarded. The program should simply show what the grade was before the adjustment and what it is afterward."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 7**, including:

- Variables and constants
- Input/output
- Formatted output
- Selection statements (if and switch)
- Loops
- Functions
- Local and global variables
- **Pointers (introduction)**

Do **not** use C language features that have not yet been introduced in the course.